Online Resource 1

Supplementary material for

Marine heatwave impact assessment on marine fauna and fisheries: a methodological synthesis and a simulation based evaluation of modelling frameworks

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1 Literature synthesis: supporting figures

1.1 Study selection

Figure S1 gives the PRISMA flow. Initial searches returned 4,869 records (Web of Science 2,182; Scopus 1,674; ScienceDirect 1,013), of which 1,559 were duplicates (45 identified manually, 1,514 by Covidence). Title, abstract and keyword screening of the remaining 3,310 records removed 2,089 as irrelevant (physical processes only; reviews, qualitative or descriptive pieces; not marine or estuarine). Of the 1,221 full texts assessed, 138 were excluded (not in English; habitat or plant studies; inaccessible), leaving 1,083 studies for the contextual synthesis, of which 76 met the additional criteria for the MHW specific analytical subset.

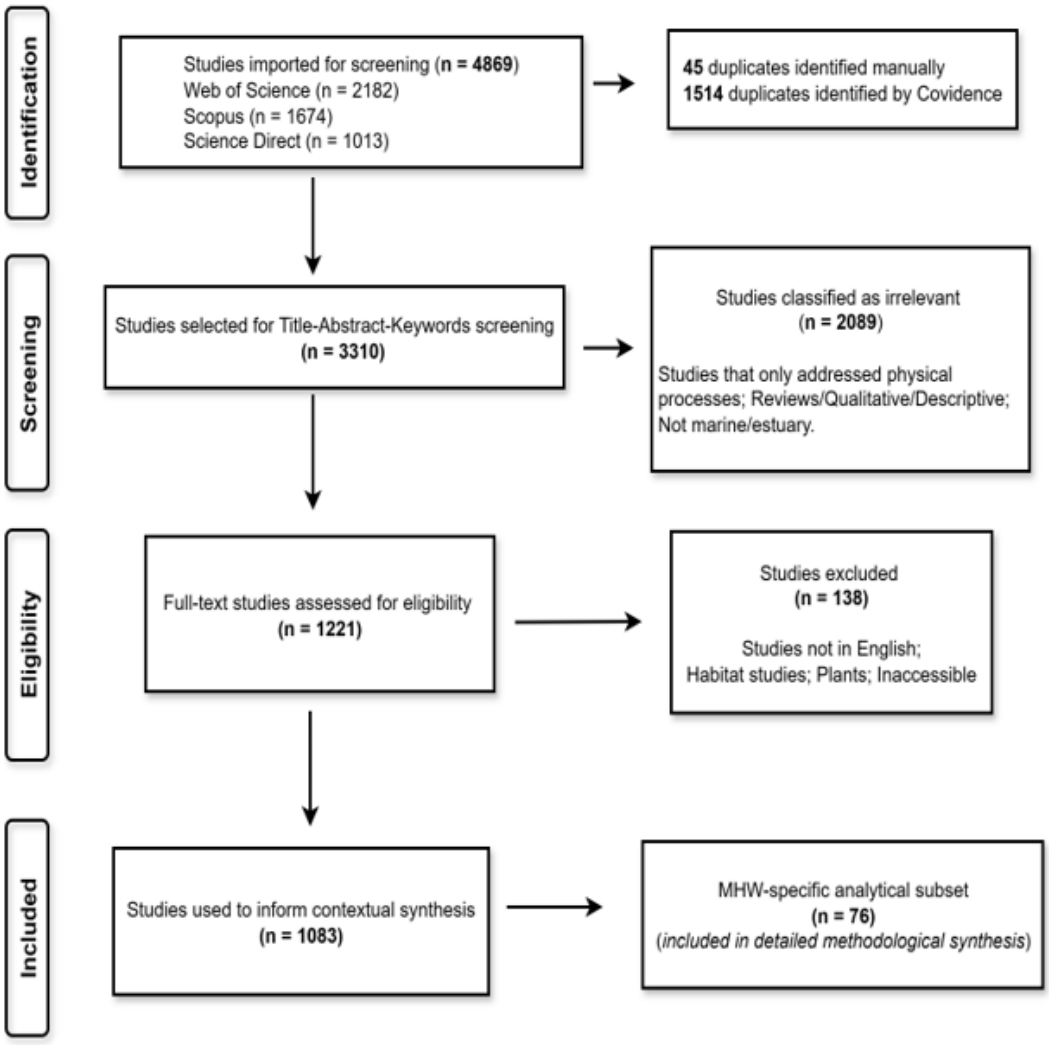


Fig. S1 PRISMA flow diagram for the literature search, screening and inclusion. Left column, records retained at each stage; right column, records removed and the reason.

1.2 Temporal dynamics of the literature

MHW specific impact research grew steadily from 2016, peaked in 2021 with 18 studies, and has remained at 11 to 13 studies a year since (Fig. S2a). Citations peaked in 2021 (601) and have declined for more recent years, as expected from citation lag. The growth coincided with wider availability of satellite derived sea surface temperature and with standardised event detection software, which lowered the technical barrier to linking exposure to biological responses. Across response families,

Population/Demography and Physiology/Bioenergetics drove the growth, expanding from minimal activity in 2016 to 2017 to peaks in 2021 before declining; Distribution/Range/Phenology followed a more moderate trajectory peaking around 2021 to 2022; Community/Ecosystem studies remained low and Fisheries/Economics endpoints were rarely addressed (Fig. S2b). The 2021 peak followed the 2014 to 2016 Northeast Pacific event, the 2015 to 2016 Tasman Sea event and the 2016 Great Barrier Reef bleaching. Whether the subsequent decline reflects an event driven research cycle or consolidation around established methods cannot be resolved from counts alone.

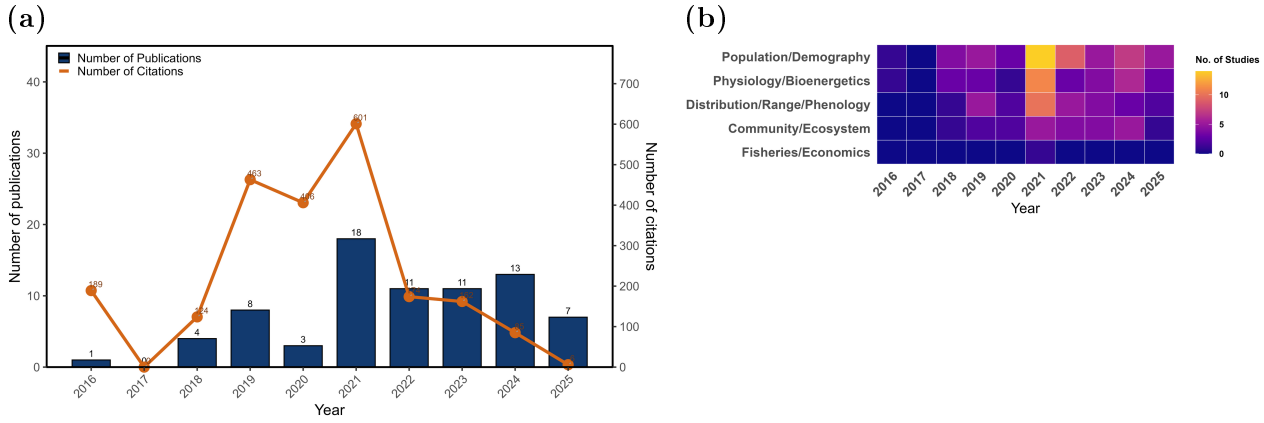


Fig. S2 Temporal dynamics of the MHW impact literature. (a) Annual number of publications (bars) and citations received (line), 2016 to 2025. (b) Number of studies by publication year and response family.

1.3 Collaboration structure

Collaboration is organised around a small number of hubs (Fig. S3a). NOAA and the University of Western Australia are the most central institutions, each embedded in a dense regional cluster: the University of Tasmania, the Australian Institute of Marine Science, CSIRO and Curtin University form a coherent Australian hub, while NOAA anchors an extensive North American network of universities and federal agencies. CNRS (France), Fisheries and Oceans Canada and IMEDEA (Spain) act as secondary connectors between otherwise distinct clusters. At national scale the United States is the principal connector, with Australia, Canada, Spain and France as secondary hubs and China, Switzerland and Portugal showing substantial international linkages (Fig. S3b). The network is sparse beyond the hubs, which is consistent with the slow diffusion of specialised methods into regions without sustained institutional capacity.

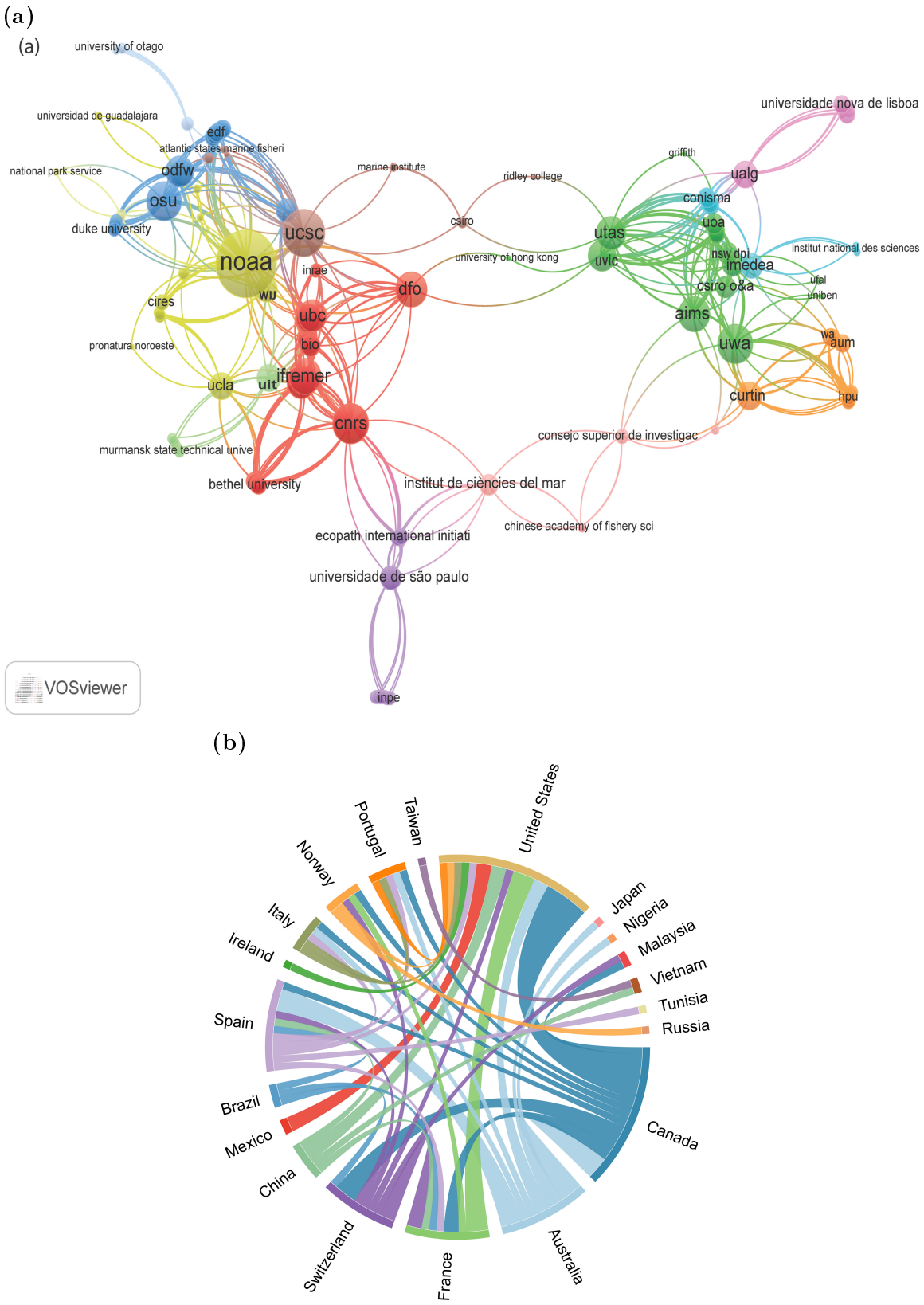


Fig. S3 Collaboration landscape of MHW impact studies. (a) Institutional co authorship network drawn in VOSviewer; node size is publication output, edge thickness is collaboration strength and colour identifies the cluster. (b) International co authorship as a chord diagram; ribbon width is the number of shared studies. Institutional acronyms are defined in Table S3.

2 Literature search and classification

2.1 Search strings

Table S1 reproduces the search string run in each of the three primary databases. In Web of Science and Scopus six term groups were combined with AND: marine heatwave terminology (including the broader terms ocean warming and climate change), a marine scope term, impact descriptors, biological endpoints, the taxonomic term fish*, and methodological terms; terms within a group were combined with OR and truncated. Publication years were restricted to 2010 to 2025 through the PY field in Web of Science and PUBYEAR limits in Scopus. ScienceDirect, whose query syntax does not support truncation or long nested Boolean strings, was searched with a reduced string of four groups. Google Scholar and NOAA repositories were searched with the core terms as supplementary sources and were not counted in the record totals. Title and abstract screening was conducted independently by three reviewers and full text assessment by two; conflicts at either stage were resolved by consensus through discussion with the wider author group and re evaluation against the inclusion criteria. Record counts are those at retrieval; 1,559 duplicates were removed before screening (Fig. S1).

Table S1 Search strings applied to each database, reproduced verbatim, with the number of records retrieved. The asterisk denotes truncation; TS, topic field (Web of Science); PY, publication year.

Database	Search string	Records
Web of Science	TS=("marine heatwave*" OR "ocean heatwave*" OR "marine heat event*" OR "extreme ocean temperature*" OR "climate change" OR "ocean warming") AND TS=(marine) AND TS=(impact* OR effect* OR response* OR "ecological consequence*" OR assessment OR sensitivity OR vulnerability) AND TS=(mortality OR growth OR abundance OR biomass OR physiology OR behavio* OR behavior* OR recruitment OR distribution OR reproduction OR survival) AND TS=(fish*) AND TS=(regression OR model* OR "data analy*" OR statistic* OR "field study" OR survey) AND PY=(2010-2025)	2,182
Scopus	(TITLE-ABS-KEY ("marine heatwave*" OR "ocean heatwave*" OR "marine heat event*" OR "extreme ocean temperature*" OR "climate change" OR "ocean warming")) AND PUBYEAR > 2009 AND PUBYEAR < 2026 AND (TITLE-ABS-KEY (marine)) AND (TITLE-ABS-KEY (impact* OR effect* OR response* OR "ecological consequence*" OR assessment OR sensitivity OR vulnerability)) AND (TITLE-ABS-KEY (mortality OR growth OR abundance OR biomass OR physiology OR behavio* OR behavior* OR recruitment OR distribution OR reproduction OR survival)) AND (TITLE-ABS-KEY (fish*)) AND (TITLE-ABS-KEY (regression OR model* OR "data analy*" OR statistic* OR "field study" OR survey))	1,674
ScienceDirect	("marine heatwave" OR "ocean warming" OR "climate change") AND marine AND (impact OR response OR effect) AND (fisheries OR fish)	1,013
Total before de duplication		4,869

2.2 Response term mapping

Free text descriptors in each study's response field were mapped to a controlled vocabulary of sub metrics and then to five response families (Population/Demography, Physiology/Bioenergetics, Distribution/Range/Phenology, Community/Ecosystem, Fisheries/Economics) with the curated regular expressions in Table S2. Response text was pre processed (lower cased, Unicode normalised, punctuation standardised) before matching. A study measuring several responses could match several sub metrics; disambiguation rules resolved the recurring ambiguities, in particular fishery dependent abundance (CPUE, landings, effort) against economic outcomes, and spatial behaviour (home range, space use) against non spatial behaviour (activity, foraging). The rule set is deposited with the repository as `data/literature/response_term_mapping.csv` and the table is typeset from that file, so every assignment can be re run.

Table S2 Regex patterns for mapping response terms to standardised submetrics and response families. Raw terms are those extracted from the studies' response fields; patterns were applied case insensitively.

Raw terms	Matched regex	Submetric	Family
Biomass; biomass changes; biomass decline; ecosystem biomass decline; spawning biomass; Trophic-Level-specific sensitivity; trophic response; Fish biomass	`biomass( changes)?|trophic(-| )level | trophic response | productivity | ecosystem biomass decline | recovery times | survey biomass distribution`	Biomass/ Productivity	Community/ Ecosystem
Community composition; Diversity; diversity changes; diversity indices; fish assemblage; functional group response; functional group shifts; functional richness; richness; species assemblage; species richness	`Diversity | richness | evenness | functional (richness | group |response) | community composition | assemblage`	Diversity/ Composition	
Asynchrony; Community shifts; community shifts; community stability; Deborealization signals; food web restructuring; resilience; resilience under protection; Tropicalization patterns; Tropicalization signals	`stability|asynchrony| resilience|tropicalization| deborealization|community shifts | food web restructuring`	Stability/ Function	
distribution; distribution shifts; Distribution; area occupied; Centre of mass depth; Centroids of distribution in depth; depth distribution; Depth of occurrence; distribution patterns; Distribution shift; distributions; habitat distribution shifts; latitude; longitude; redistribution Shifts in center of gravity; size distribution; Species-specific spatial distribution; survey biomass distribution under scenarios; Vertical distribution	`Distribution ( shifts | patterns)? | redistribution | centroid |center of gravity | area occupied | latitude | longitude | vertical distribution | depth distribution | centre of mass depth | mesopelagic patch size | depth of occurrence`	Distribution (2D/Depth)	Distribution/ Range/ Phenology
habitat suitability; habitat loss impacts; habitat quality; Habitat suitability; habitat use; port availability; projected habitat suitability	`habitat (use | quality | loss) | habitat suitability | projected habitat suitability | environmental habitat associations | port availability`	Habitat/ Habitat Suitability	
Migration routes	`migration routes | home range | space use | diel overlap`	Movement/ Space use	
Phenology; phenology; phenology shifts; spawning shifts	`Phenology | hatch timing | spawning shifts | timing`	Phenology	
biogeographic shifts; Poleward range shifts; range expansion; range shift; Range shift persistence	`range shift |range expansion | biogeographic shifts |poleward range`	Range change	
food security risks; revenue shocks	`Revenue | profit | price(s)? | income | food security | employment | cost(s)? | market | economic | livelihood`	Revenue/ Markets/Food security	Fisheries/ Economics
Activity level; behaviour; diel behavior; grazing; swimming performance	`activity level | behaviour | foraging | grazing | swimming performance | avoidance | boldness | diel`	Behaviour/ Performance	Physiology/ Bioenergetics
energy budget; feed efficiency; Flesh colour; Nutritional quality; tissue quality	`energy budget | feed efficiency | flesh colour | tissue quality | nutritional quality`	Bioenergetics/ Quality	

continued on next page

Table S2 (continued)

Raw terms	Matched regex	Submetric	Family
age-0 body condition; body condition; Residual Condition Index of selected fish	`body condition | condition index | residual condition`	Condition	
growth; Growth; growth patterns; growth performance; Growth rates; growth trajectories; relative growth rate	`Growth (rate | trajectories | patterns)? | relative growth rate`	Growth	
antioxidant activity; immune health; immune responses; lysozyme	`Immune | igm | lysozyme | antioxidant`	Immunity	
Aerobic metabolism; excretion; metabolic rate; metabolism; oxygen consumption; respiration; routine metabolic rate	`metaboli(c|sm) | respiration | excretion | oxygen consumption | routine metabolic`	Metabolism	
DNA methylation patterns (DMRs); environmental habitat associations; gene expression; gene expression in tissues; home range shifts; mesopelagic patch size; metabolomic pathways; metastable epialleles (MEs); mitigation from mesophotic refuges; recovery times; transcriptome	`gene expression | transcriptome | dna methylation | epigenetic | dmr | me(s)?`	Molecular/ Epigenetic	
acclimation capacity; CTmax; Physiological stress; thermal tolerance thresholds	`Ctmax | thermal tolerance | physiological stress | acclimation capacity`	Thermal/ Stress Tolerance	
CPUE; Catch; Bycatch rates; catch; Catch and Effort; CPUE catchability; effort; fleet-specific catchability shifts; Length-at-catch distribution; maximum catch potential; Probability of occurrence by fleet; Spatial catch	`Cpue | catchability |catch(,| ;|$| ) | effort | landings | bycatch | fleet`	Abundance/ CPUE (fishery-dependent)	Population/ Demography
Abundance; abundance; abundance shifts; Abundance (CPUE); densities; density; density changes; Fish abundance; fish densities; juvenile abundance; larval abundance; Species abundance; standardized abundance; temperature & abundance relationships; Urchin density	`Abundance | juvenile abundance | standardized abundance | densit(y|ies)`	Abundance/ Density	
survival; Survival; mortality; Mortality thresholds; Calf mortality; Embryo survival; extirpation risk; potential selective mortality; survival under harvesting + MHW stress	`Mortality |extirpation risk | survival(?!.*post)`	Mortality/ Survival	
recruitment; recruitment failure; recruitment failure mechanisms; recruitment indeces; recruitment index; recruitment patterns; recruitment success; Recruitment variation	`Recruitment (index | failure | patterns | success | variation)?`	Recruitment	
Calf production; Egg buoyancy; hatch timing; Hatch timing; pup numbers; reproductive output; spawning stock biomass	`reproductive output | egg | larval | hatch | calf | pup | spawning stock biomass | embryo`	Reproduction/ Early life	
length frequency; population structure; size structure; size structure (length classes)	`size structure | length (frequency | classes) | population structure | age structure`	Size/Age Structure	

2.3 Institutional acronyms

Table S3 Institutional acronyms used in Fig. S3.

Acronym	Institution
AIMS	Australian Institute of Marine Science
CNRS	Centre National de la Recherche Scientifique, France
CSIRO	Commonwealth Scientific and Industrial Research Organisation, Australia
DFO	Fisheries and Oceans Canada
IMEDEA	Mediterranean Institute for Advanced Studies, Spain
NOAA	National Oceanic and Atmospheric Administration, United States
UCSC	University of California, Santa Cruz
UTAS	University of Tasmania
UWA	University of Western Australia

3 Detection verification and sensitivity

3.1 Reference values

Table S4 gives the characteristics of the most intense event (by peak anomaly) in each of the three real OISST series under the Hobday defaults, as computed by heatwaveR 1.0.0 on R 4.3.3. These are the reference values against which the repository's test suite checks; they are regenerated on the analyst's machine by `scripts/verify_detection.R`.

Table S4 Most intense event by peak anomaly in each OISST series under the Hobday defaults (1983 to 2012 climatology, 90th percentile, 5 day minimum, 2 day gap). Intensities in °C relative to climatology; cumulative intensity in °C days.

Region	Start	End	Days	Max	Mean	Cumulative	Events in series
Western Australia	2011-02-06	2011-04-06	60	6.51	3.21	192.7	76
Northwest Atlantic	2020-06-22	2020-07-03	12	5.19	2.78	33.4	114
Mediterranean	2017-06-10	2017-06-27	18	5.51	3.61	64.9	119

3.2 Sensitivity of the 2011 Western Australian event

Table S5 shows how the 2011 event's characteristics change across three climatology windows and two percentile thresholds. The event is robust to the 95th percentile and to extending the window to 2022, but under a 1982 to 2011 window at the 90th percentile the detection algorithm joins the 2011 anomaly to a different event structure and the largest 2011 event by peak anomaly becomes a short November event. This is the sensitivity that item 2 of the reporting standards asks authors to disclose.

Table S5 Largest 2011 event by peak anomaly in the Western Australian series under alternative climatology windows and percentile thresholds.

Climatology	Percentile	Start	End	Days	Max	Mean	Cumulative
1982-01-01 to 2011-12-31	90	2011-11-21	2011-11-26	6	2.48	1.89	11.4
1982-01-01 to 2011-12-31	95	2011-02-07	2011-04-01	54	6.58	3.48	187.8
1982-01-01 to 2022-12-31	90	2011-02-06	2011-04-01	55	6.49	3.34	183.7
1982-01-01 to 2022-12-31	95	2011-02-07	2011-03-19	41	6.49	3.89	159.4
1983-01-01 to 2012-12-31	90	2011-02-06	2011-04-06	60	6.51	3.21	192.7
1983-01-01 to 2012-12-31	95	2011-02-07	2011-03-19	41	6.51	3.90	160.1

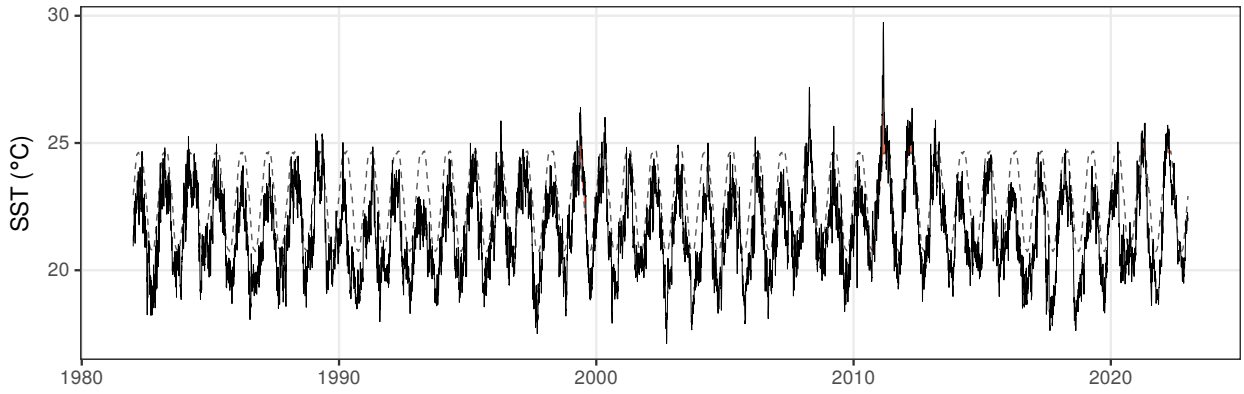


Fig. S4 Daily OISST for the Western Australian series, 1982 to 2022 (black), the 90th percentile threshold (dashed grey) and exceedance above threshold (shaded) for every detected event. The 2011 event is the tallest exceedance; the 1999 event, longer but lower, is the largest by cumulative intensity.

4 Data generating process calibration

Table S6 reports the realised response distribution for every combination of true effect and true lag on one 9 site and one 25 site panel per region (range 50 km, replicate 1). All cells are usable: no cell has more than 90% zeros, fewer than five distinct values, or an implausible mean.

Table S6 Realised response distribution by region, grid size, true effect and true lag (range 50 km, replicate 1).

Region	Sites	β_1	Lag	Zeros (%)	Mean y	Max y	Distinct	Usable
Western Australia	9	-0.04	0	15	8.0	84	36	yes
Western Australia	9	-0.04	1	15	7.8	57	33	yes
Western Australia	9	-0.04	2	12	7.9	43	33	yes
Western Australia	9	-0.08	0	24	6.5	38	30	yes
Western Australia	9	-0.08	1	20	6.7	40	33	yes
Western Australia	9	-0.08	2	22	6.7	49	30	yes
Western Australia	9	-0.12	0	31	5.9	56	35	yes
Western Australia	9	-0.12	1	25	6.3	51	32	yes
Western Australia	9	-0.12	2	25	6.6	58	32	yes
Western Australia	25	-0.04	0	14	7.9	47	43	yes
Western Australia	25	-0.04	1	13	8.3	51	40	yes
Western Australia	25	-0.04	2	13	8.5	51	41	yes
Western Australia	25	-0.08	0	23	6.5	52	43	yes
Western Australia	25	-0.08	1	21	6.7	55	39	yes
Western Australia	25	-0.08	2	20	7.2	57	42	yes
Western Australia	25	-0.12	0	30	5.5	57	39	yes
Western Australia	25	-0.12	1	28	6.1	50	37	yes
Western Australia	25	-0.12	2	26	6.3	49	37	yes
Northwest Atlantic	9	-0.04	0	31	5.9	39	32	yes
Northwest Atlantic	9	-0.04	1	26	6.6	45	34	yes
Northwest Atlantic	9	-0.04	2	26	6.4	36	31	yes
Northwest Atlantic	9	-0.08	0	39	5.3	46	34	yes
Northwest Atlantic	9	-0.08	1	38	5.0	40	32	yes
Northwest Atlantic	9	-0.08	2	34	5.5	32	30	yes
Northwest Atlantic	9	-0.12	0	44	4.1	30	28	yes
Northwest Atlantic	9	-0.12	1	42	4.6	43	29	yes
Northwest Atlantic	9	-0.12	2	41	5.0	50	31	yes

Region	Sites	β_1	Lag	Zeros (%)	Mean y	Max y	Distinct	Usable
Northwest Atlantic	25	-0.04	0	30	6.2	52	40	yes
Northwest Atlantic	25	-0.04	1	28	6.3	41	39	yes
Northwest Atlantic	25	-0.04	2	26	6.3	57	38	yes
Northwest Atlantic	25	-0.08	0	38	4.9	54	40	yes
Northwest Atlantic	25	-0.08	1	38	5.1	62	39	yes
Northwest Atlantic	25	-0.08	2	32	5.4	60	38	yes
Northwest Atlantic	25	-0.12	0	45	4.1	41	34	yes
Northwest Atlantic	25	-0.12	1	43	4.7	49	37	yes
Northwest Atlantic	25	-0.12	2	40	4.9	41	37	yes
Mediterranean	9	-0.04	0	33	5.5	35	29	yes
Mediterranean	9	-0.04	1	31	6.0	36	30	yes
Mediterranean	9	-0.04	2	28	6.1	40	33	yes
Mediterranean	9	-0.08	0	41	3.9	38	27	yes
Mediterranean	9	-0.08	1	38	4.3	31	27	yes
Mediterranean	9	-0.08	2	35	4.2	29	24	yes
Mediterranean	9	-0.12	0	44	3.2	29	24	yes
Mediterranean	9	-0.12	1	41	3.6	37	27	yes
Mediterranean	9	-0.12	2	43	3.6	38	23	yes
Mediterranean	25	-0.04	0	33	5.4	46	36	yes
Mediterranean	25	-0.04	1	31	5.7	43	40	yes
Mediterranean	25	-0.04	2	29	6.1	46	36	yes
Mediterranean	25	-0.08	0	40	4.3	36	30	yes
Mediterranean	25	-0.08	1	37	4.2	40	32	yes
Mediterranean	25	-0.08	2	35	4.7	49	35	yes
Mediterranean	25	-0.12	0	45	3.2	39	28	yes
Mediterranean	25	-0.12	1	43	3.7	47	33	yes
Mediterranean	25	-0.12	2	41	4.0	51	32	yes

5 Full simulation results

This section gives every aggregation of the simulation output on which the main text draws. All tables are generated from `results/tables/` by `scripts/make_supp_tables.R`; the raw file `R_raw_results.csv` (one row per model fit) is deposited with the repository. Model classes are described in Table 4 of the main text; the estimand reported as $\hat{\beta}_1$ differs between classes as stated there.

5.1 Model level summary

Table S7 averages over all 108 scenarios. The ordering by RMSE places the two GLMs ahead of the DLNM because their estimates, although biased, vary little across replicates; the ordering by mean bias places the DLNM first. Coverage and log score are averaged only over scenarios where the quantity is defined (no interval for the random forest; no proper predictive density for the quasi Poisson DLNM). The sdmTMB row summarises the 14 panels on which that model was run.

Table S7 Model level performance across all scenarios. Coverage is the mean empirical coverage of nominal 95% intervals over scenarios where an interval is defined; log score is the mean held out negative binomial log predictive density; Moran's I is the mean residual spatial autocorrelation of site mean residuals.

Model	Scenarios	Mean bias	Median RMSE	Coverage	Log score	Moran's I	Conv. (%)	Median s
Naive NB GLM	108	0.039	0.030	0.05	-1.28	-0.084	96.5	0.04
Lag-1 NB GLM	108	0.036	0.029	0.07	-1.38	-0.083	96.6	0.04
NB GAM	108	0.015	0.053	0.69	-1.60	-0.079	100.0	0.13
DLNM	108	-0.005	0.036	0.73	–	-0.082	100.0	0.02
sdmTMB	108	0.036	0.029	0.12	–	-0.087	4.0	21.65
Random forest	108	0.053	0.048	–	-1.34	-0.216	100.0	0.16

5.2 Bias by true lag

Table S8 is the breakdown behind Fig. 5 of the main text. The naive GLM's bias changes sign between lag 0 and lag 1 and stays at that level at lag 2; the lag-1 GLM shows the mirror image, small at lag 1 and large at lags 0 and 2; the DLNM's bias stays within a narrow band at all three lags but with a larger standard deviation across replicates.

Table S8 Bias, RMSE and interval coverage by model class and true lag, averaged over regions, effect sizes, spatial ranges and grid sizes.

Model	True lag	Mean bias	Median bias	RMSE	Coverage
Naive NB GLM	0	-0.016	-0.016	0.016	0.16
Naive NB GLM	1	0.065	0.065	0.065	0.00
Naive NB GLM	2	0.069	0.067	0.067	0.00
Lag-1 NB GLM	0	0.058	0.057	0.057	0.00
Lag-1 NB GLM	1	-0.016	-0.015	0.016	0.21
Lag-1 NB GLM	2	0.065	0.066	0.066	0.00
NB GAM	0	0.058	0.061	0.061	0.66
NB GAM	1	-0.039	-0.035	0.044	1.00
NB GAM	2	0.026	0.041	0.062	0.42
DLNM	0	0.021	0.013	0.033	0.74
DLNM	1	-0.031	-0.039	0.051	0.54
DLNM	2	-0.004	-0.003	0.027	0.92
sdmTMB	0	0.053	0.057	0.057	0.11
sdmTMB	1	-0.015	-0.016	0.016	0.25
sdmTMB	2	0.071	0.066	0.066	0.00
Random forest	0	0.080	0.081	0.082	–
Random forest	1	0.007	0.008	0.030	–
Random forest	2	0.071	0.069	0.081	–

5.3 Relative bias by true effect and lag

Table S9 expresses the same biases as a percentage of the true coefficient (Fig. 6 of the main text). Because the absolute bias under lag misspecification is approximately constant, the percentage is largest for the smallest true effect ($\beta_1 = -0.04$), where the naive GLM's estimate can exceed the true magnitude in the wrong direction at lag 0 and vanish entirely at lags 1 and 2.

Table S9 Relative bias (%) of the estimated exposure coefficient, $100(\hat{\beta}_1 - \beta_1)/\beta_1$, by model class, true effect and true lag. Negative values indicate under estimation of the magnitude of a negative effect.

Model	β_1	Lag 0	Lag 1	Lag 2
Naive NB GLM	-0.12	10	-83	-89
Naive NB GLM	-0.08	20	-81	-86
Naive NB GLM	-0.04	53	-74	-77
Lag-1 NB GLM	-0.12	-77	10	-83
Lag-1 NB GLM	-0.08	-72	19	-81
Lag-1 NB GLM	-0.04	-59	52	-73
NB GAM	-0.12	-78	26	-35
NB GAM	-0.08	-73	47	-37
NB GAM	-0.04	-57	115	-12
DLNM	-0.12	-21	25	-5
DLNM	-0.08	-25	42	12
DLNM	-0.04	-45	69	25
sdmTMB	-0.12	-68	7	-92
sdmTMB	-0.08	-67	20	-87
sdmTMB	-0.04	-51	50	-84
Random forest	-0.12	-100	-14	-97
Random forest	-0.08	-99	-12	-90
Random forest	-0.04	-99	11	-62

5.4 Bias by region

Table S10 is the breakdown behind Fig. 9 of the main text. Regional exposure distributions differ substantially (mean annual cumulative intensity of roughly 20, 50 and 40 °C days for the Western Australian, Northwest Atlantic and Mediterranean series; Table S6), yet the bias of each model class is nearly constant across regions, which is what is expected when the bias arises from the model's temporal assumption rather than from the exposure values.

Table S10 Bias, RMSE and coverage by model class and region.

Model	Region	Mean bias	RMSE	Coverage
Naive NB GLM	Mediterranean	0.038	0.029	0.07
Naive NB GLM	Northwest Atlantic	0.036	0.030	0.03
Naive NB GLM	Western Australia	0.043	0.038	0.06
Lag-1 NB GLM	Mediterranean	0.035	0.029	0.08
Lag-1 NB GLM	Northwest Atlantic	0.034	0.029	0.06
Lag-1 NB GLM	Western Australia	0.038	0.030	0.07
NB GAM	Mediterranean	0.028	0.050	0.66
NB GAM	Northwest Atlantic	-0.012	0.059	1.00
NB GAM	Western Australia	0.029	0.038	0.41
DLNM	Mediterranean	0.012	0.030	0.93
DLNM	Northwest Atlantic	-0.017	0.031	0.77
DLNM	Western Australia	-0.009	0.058	0.51
sdmTMB	Mediterranean	0.029	0.028	0.17
sdmTMB	Northwest Atlantic	0.039	0.031	0.12
sdmTMB	Western Australia	0.038	0.037	0.07
Random forest	Mediterranean	0.051	0.040	–
Random forest	Northwest Atlantic	0.051	0.053	–
Random forest	Western Australia	0.057	0.044	–

5.5 Bias by grid size and spatial range

Table S11 splits the results by the two spatial design factors. Neither factor changes the bias of any class detectably. The number of sites changes precision (RMSE falls from 9 to 25 sites for the

classes whose estimates are unbiased at the correct lag), and the correlation range changes nothing, because the site effects absorb the correlated component of the exposure noise.

Table S11 Bias, RMSE and coverage by model class, grid size and spatial correlation range.

Model	Factor level	Mean bias	RMSE	Coverage
Naive NB GLM	25 sites	0.039	0.030	0.03
Naive NB GLM	9 sites	0.039	0.030	0.08
Naive NB GLM	range 0 km	0.039	0.030	0.06
Naive NB GLM	range 50 km	0.039	0.030	0.05
Lag-1 NB GLM	25 sites	0.036	0.029	0.03
Lag-1 NB GLM	9 sites	0.035	0.029	0.11
Lag-1 NB GLM	range 0 km	0.036	0.029	0.07
Lag-1 NB GLM	range 50 km	0.036	0.029	0.07
NB GAM	25 sites	0.009	0.054	0.69
NB GAM	9 sites	0.021	0.052	0.69
NB GAM	range 0 km	0.015	0.051	0.69
NB GAM	range 50 km	0.015	0.054	0.69
DLNM	25 sites	-0.003	0.029	0.64
DLNM	9 sites	-0.006	0.043	0.82
DLNM	range 0 km	-0.005	0.041	0.73
DLNM	range 50 km	-0.004	0.035	0.74
sdmTMB	25 sites	0.038	0.035	0.08
sdmTMB	9 sites	0.033	0.028	0.16
sdmTMB	range 0 km	0.036	0.030	0.17
sdmTMB	range 50 km	0.035	0.029	0.07
Random forest	25 sites	0.057	0.046	–
Random forest	9 sites	0.049	0.052	–
Random forest	range 0 km	0.052	0.049	–
Random forest	range 50 km	0.054	0.048	–

5.6 Convergence and runtime

Table S12 is the run report printed by `scripts/aggregate.R`. The percentage returning an estimate and the percentage passing the package's own convergence check are reported separately: a fit that triggered a warning (for the GLMs, typically the theta iteration limit in `glm.nb`; for sdmTMB, a non positive definite Hessian reported by `sanity()`) was retained with its estimate and counted as not converged, rather than being dropped. Median runtime per fit is on one core; package versions are given in Section 3.3 of the main text and the full `sessionInfo()` output is deposited with the repository as `results/tables/sessionInfo.txt`.

Table S12 Run report: number of fits, percentage returning a point estimate, percentage returning an interval, percentage passing the package's convergence check, and median wall clock time per fit.

Model	Fits	Estimate (%)	Interval (%)	Converged (%)	Median s
Naive NB GLM	1080	99.9	99.9	96.5	0.04
Lag-1 NB GLM	1080	99.9	99.9	96.6	0.04
NB GAM	1080	100.0	100.0	100.0	0.14
DLNM	1080	100.0	99.7	100.0	0.02
sdmTMB	126	96.8	78.6	4.0	21.65
Random forest	1080	100.0	0.0	100.0	0.18

5.7 Additional figures

Figures S5 and S6 complement the main text figures. The RMSE figure shows the precision cost of the DLNM's robustness: its boxes are wider than the correctly lagged GLM's at every lag. The Moran's

I figure shows that residual spatial autocorrelation is small and slightly negative for every regression class on both independent and correlated panels, which confirms that the correlated exposure noise is absorbed by the site effects and leaves no spatial structure for a random field model to explain.

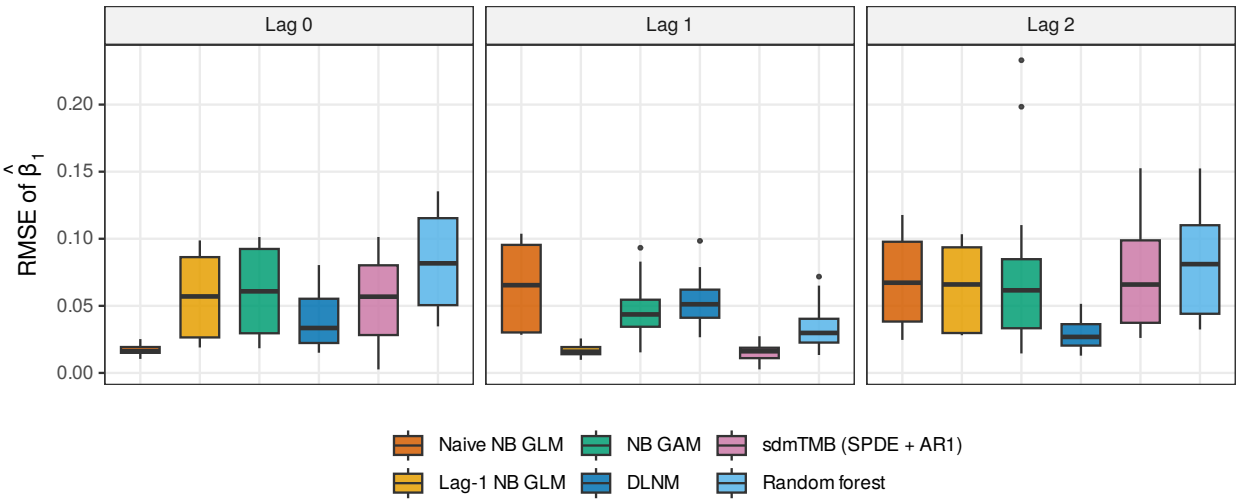


Fig. S5 Root mean squared error of the estimated exposure coefficient by model class and true lag; each box summarises one third of the 108 scenarios. Colours identify model classes consistently with the main text.

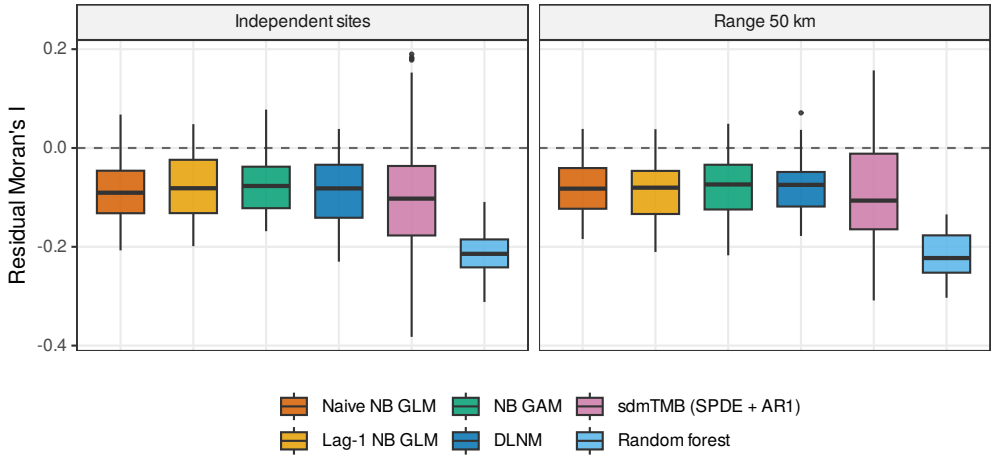


Fig. S6 Residual Moran's I of site mean residuals by model class, for spatially independent panels and panels with a 50 km correlation range.

6 Repository structure and reproduction

```
mhw-R-audit/
  R/00_config.R design constants, seeds, paths
  R/01_detection.R heatwaveR wrappers
  R/02_exposure.R annual metrics, lag panels
  R/03_panel.R OISST loader, Matern spatial panel
  R/04_dgp.R response simulator, degeneracy checks
  R/05_models.R six fitters, unified return interface
  R/06_runner.R resumable checkpointed runner
  R/07_aggregate.R summaries, run report
  scripts/verify_detection.R RUN FIRST: reference values
  scripts/diagnose_dgp.R RUN SECOND: calibration check
  scripts/run_chunk.R one array task
```

```
  scripts/aggregate.R summaries
  scripts/make_figures.R figures
  scripts/make_results_macros.R manuscript macros
  scripts/make_main_tables.R main text tables
  scripts/make_supp_tables.R supplement tables
  scripts/make_literature_tables.R Tables S1 and S2 from data/literature
  scripts/prepare_literature_figures.R stage the synthesis figure panels
  data/literature/search_strategy.csv search strings (Table S1)
  data/literature/response_term_mapping.csv response term rules (Table S2)
  hpc/slurm_array.sh, hpc/pbs_array.sh, hpc/run_local_parallel.sh
  tests/testthat/test-detection.R
  manuscript/main.tex, supplement.tex, refs.bib
```

Reproduction on a workstation:

```
export MHW_ROOT=$PWD
Rscript install/install_packages.R --all
Rscript scripts/verify_detection.R
Rscript scripts/diagnose_dgp.R
Rscript -e 'library(testthat); test_dir("tests/testthat")'
bash hpc/run_local_parallel.sh 8
Rscript scripts/make_results_macros.R
cd manuscript && latexmk -pdf main.tex supplement.tex
```

7 Model fitter interface

Every fitter in `R/05_models.R` returns the same fields, so that a new model can be added without changing the runner:

```
fit_<model>(d) # d: y, site, year, X, Y, i_cum, i_cum_lag0..3, mhw_days
# returns list(model_id, beta1_hat, beta1_lo, beta1_hi,
# log_score, morans_I, converged, runtime_sec, note)
```

The sdmTMB fitter illustrates the convention for convergence reporting: warnings are muffled rather than treated as failure, and the package's own `sanity()` verdict is recorded so that flagged fits enter the run report rather than vanishing as missing rows.

```
m <- withCallingHandlers(
  sdmTMB::sdmTMB(y ~ i_cum_lag1, data = tr, mesh = mesh, time = "year",
                 family = sdmTMB::nbinom2(link = "log"),
                 spatial = "on", spatiotemporal = "ar1", silent = TRUE),
  warning = function(w) invokeRestart("muffleWarning"))
converged <- isTRUE(sdmTMB::sanity(m, silent = TRUE)$all_ok)
```

References